
Post-Implementation Power Optimization of an Operand-Gated ALU using FPGA Fabric

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High-activity workloads in parallel computing blocks represent a primary driver of dynamic power dissipation in modern digital design due to redundant transistors switching. This project addresses the challenge of parasitic power waste in deep-bus arithmetic architectures by designing and validating a 32-bit hierarchical ALU optimised through input-isolation operand gating. While a standard ALU concurrently ripples data across all internal units, the developed architecture strategically clamps data paths to idle functional blocks based on immediate opcode decoding, forcing local switching activity factors to zero. To verify real-world performance, a post-implementation physical signoff methodology was used, with functional simulation stress tests that injected an annotated Switching Activity Interchange Format (SAIF) matrix back into the physical netlist. Results demonstrate that the operand-gated architecture achieves a notable 17.65% reduction in the physical fabric's dynamic power under high-stress conditions compared to the baseline design running at 50 MHz.

1 Theoretical Foundations of Operand-Gated Architectures

1.1 Dynamic Power Formulation

For decades, Dennard scaling governed the advancement of digital CMOS circuits, stating that as transistors shrank, their power density remained constant due to a proportional reduction in supply voltage. However, sub-micron scaling has hit physical boundaries, dominated by short-channel effects and exponential gate-oxide leakage currents. Hence, supply voltage scaling has flattened, causing modern architectures to operate under strict thermal design power budgets.

To analyze and mitigate these thermal constraints, the fundamental physical behavior of a CMOS circuit must be evaluated through the classic dynamic power dissipation equation:

$$P_{\text{dynamic}} = \alpha \cdot C_{\text{eff}} \cdot V_{dd}^2 \cdot f \quad (1)$$

Where:

- α represents the switching activity factor ($0 \leq \alpha \leq 1$), defining the average number of power-consuming logic transitions per clock cycle.

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- C_{eff} represents the total parasitic load capacitance of the physical routing traces and transistor gates.
 - V_{dd} represents the operating supply voltage.
 - f represents the clock frequency.

Because physical hardware constraints limit the reduction of V_{dd} , and scaling down the clock frequency (f) directly penalises system throughput, modern hardware optimization must explicitly target structural variables. While minimizing layout area can lower C_{eff} , this project aims to directly optimize the *switching activity factor* (α). By eliminating unnecessary electrical transitions in idle silicon domains, dynamic power can be suppressed at the architectural level without degrading the physical manufacturing process or reducing clock speeds.

1.2 Standard ALU Architectures

In a conventional hierarchical 32-bit ALU, input operands are distributed globally to all functional processing blocks concurrently via shared input buses.

When new data packages arrive at the primary input ports, the electrical signals immediately ripple through every sub-module simultaneously, regardless of the selected instruction. For instance, even if the system executes a simple bitwise AND operation, the deep logic gates of the 32-bit Multiplier still compute, charge, and discharge internal parasitic capacitances as the inputs fluctuate.

This continuous, toggling across unselected logic trees causes severe dynamic power dissipation. Transistors consume energy to compute discarded results, yielding an artificially high global switching activity factor (α).

1.3 Input Isolation Theory

The core microarchitectural solution implemented in this project is *Operand Gating*. The objective is to decouple idle blocks from active data networks dynamically. By monitoring the incoming opcode, the system determines exactly which execution path is required for the current clock cycle and isolates the remaining sub-modules. Front-end gating elements, such as transparent multiplexers or bitwise masking logic arrays, are inserted directly onto the input boundaries of the sub-modules.

The mathematical mechanics of operand gating can be modelled by analyzing its direct impact on local switching activity factors. Consider an ALU containing N independent execution blocks. In a standard architecture, the global switching factor is a summation of all active internal logic nodes:

$$\alpha_{\text{global}} = \sum_{i=1}^N \alpha_i \quad (2)$$

Under the proposed operand gating scheme, the input isolation cells function as a mathematical filtering envelope controlled by a discrete decoding function $g_i(\text{opcode}) \in \{0, 1\}$. When a module i is unselected, the control loop forces its input lines to hold a fixed,

unchanging static logic value (e.g., all zeros). Hence, the local transition factor drops to absolute zero:

$$\alpha_{i,\text{gated}} = \alpha_i \cdot g_i(\text{opcode}) \implies \text{if } g_i(\text{opcode}) = 0, \alpha_{i,\text{gated}} = 0 \quad (3)$$

By freezing the local toggle rate within the unselected blocks, the internal transistors remain static, completely neutralizing dynamic power consumption inside that sector of the silicon fabric. However, this microarchitectural advantage introduces a prominent hardware trade-off. The addition of front-end gating cells places extra physical logic gates directly along the input paths. These structures add a definitive layer of propagation delay (t_{pd}), which reduces the design’s timing margin and decreases its maximum stable operational frequency ($F_{\max}$).

2 Microarchitectural Hardware Design (RTL)

2.1 Structural Design Hierarchy

To evaluate the power characteristics of operand gating, a highly modular, decoupled microarchitecture was designed and implemented. The design partitions processing responsibilities across five structural files to isolate functional domains.

- **alu_pkg.vhd:** A centralized design library containing global constants, bus width definitions (`DATA_WIDTH = 32`, `OP_WIDTH = 4`), and the enumerated opcode matrix for all 16 arithmetic, logic, and shifting operations.
- **arith_unit.vhd:** The arithmetic execution engine responsible for mathematical computations. This block encapsulates 32-bit addition, subtraction, and a physical 32-bit combinational unsigned multiplier tree.
- **logic_unit.vhd:** A purely bitwise execution block performing Boolean functions (AND, OR, XOR, NOT) across the 32-bit operand inputs.
- **shift_unit.vhd:** A dedicated barrel shifter capable of multi-bit arithmetic and logical left/right shifting operations, implemented via a multi-stage combinational multiplexer fabric.
- **top_alu.vhd:** The structural top-level wrapper that instantiates the sub-modules and establishes the internal signal interconnect network. In the low-power variant of the design, this entity embeds the front-end operand-gating multiplexer arrays and control decoding loop.

2.2 Front-End Filtering

In the standard ALU, input operands are permanently mapped directly to the inputs of all execution blocks. In contrast, the low-power ALU architecture intercepts incoming data packages by routing them through a front-end filtering barrier within the `top_alu.vhd` wrapper.

The gating control loop operates on a pure look-ahead combinatorial decoding scheme. When operands arrive at the primary *src_A* and *src_B* ports, they do not immediately cross into the sub-modules. Instead, the 4-bit *opcode* bus is decoded concurrently within a tracking control process. Based on the decoded value, a set of internal select vectors activates exactly one processing domain while clamping the inputs of the remaining two dormant modules to a static ground state (`x"00000000"`).

For example:

$$\text{If } opcode \in \text{Logic Class} \implies \begin{cases} Inputs_{Logic_Unit} = src_A, src_B \\ Inputs_{Arith_Unit} = \text{x"00000000"} \\ Inputs_{Shift_Unit} = \text{x"00000000"} \end{cases} \quad (4)$$

By acting as an architectural filter, the gating loop prevents changing data bits from reaching the massive capacitance nets of the unselected hardware. Because the inputs to other sub-modules remain perfectly static during logical operations, no charging or discharging occurs across their internal nodes, confining dynamic switching power strictly to the logic unit.

3 Simulation-Driven Verification Methodology

3.1 The High Stress Test Workload

The accuracy of a post-implementation power analysis is heavily dependent on the quality of the input stimulus applied during verification. In many introductory digital design workflows, testbenches rely on basic binary counters to drive input data buses. While a counter-driven stimulus is sufficient for verifying basic functional correctness, it introduces extreme distortions when used for power estimation. In a 32-bit binary counter, the least significant bit (LSB) toggles on every clock edge, but the upper bits toggle exponentially slower ($f/2^n$). This leaves the upper data lines practically static for large windows of time, artificially masking the real power footprint of the design by suppressing the switching activity factor (α).

To eliminate this measurement bias and evaluate the ALU under a realistic operating scenario, the testbench (`tb_top_alu.vhd`) implements a custom *High Stress Test Workload*.

The stimulus architecture operates via a dual-loop matrix:

1. **Opcode Sweep:** An inner loop iterates sequentially through all 16 independent arithmetic, logical, and shifting opcodes (from 0 to 15). This ensures that every functional domain inside the ALU is stimulated back-to-back.
2. **Workload Iteration:** An outer loop repeats this entire 16-operation sweep 10 times consecutively, establishing a prolonged, high-activity simulation timeline totaling 3,300 ns.

To force the highest possible switching frequency, the input variable data arrays are structurally manipulated on every single instruction pass. Rather than incrementing monotonically, the testbench variables undergo a complete bitwise inversion on every clock

cycle, alternating back-to-back between `x"55555555"` (binary 0101...) and `x"AAAAAAAA"` (binary 1010...).

$$\text{Cycle } n : \begin{cases} src_A = x"55555555" \\ src_B = x"AAAAAAAA" \end{cases} \implies \text{Cycle } n + 1 : \begin{cases} src_A = \text{NOT } src_A \\ src_B = \text{NOT } src_B \end{cases} \quad (5)$$

This aggressive inversion forces all 32 bits of both primary input buses (`src_A` and `src_B`) to flip simultaneously on every step. This stress test provides a rigorous baseline that exposes the efficiency of the front-end operand-gating isolation multiplexers.

3.2 Post-Implementation SAIF Generation

To transition the power analysis from an approximation into a robust verification, a deterministic tool flow was developed. Rather than relying on Vivado’s default Vectorless Estimation, this project records the explicit physical switching behavior of the layout via a *Switching Activity Interchange Format (SAIF)* pipeline.

The execution pipeline transitions systematically through three states:

1. **Physical Placement and Routing:** The synthesized VHDL netlist is fully implemented on the target Artix-7 silicon die, generating a timing-accurate physical gate representation where all internal signal traces are bounded to real hardware coordinates.
2. **Post-Implementation Functional Simulation:** The physical netlist is loaded into the Vivado simulator alongside the testbench. This configuration mimics the exact cycle-by-cycle electrical behavior of a manufactured chip.
3. **SAIF Extraction:** Before running the simulation, the Tcl console initializes a logging instance targeting the hierarchical Unit Under Test (UUT). The specific operational sequence is executed as follows:

```
restart
open_saif "SAIF_file_directory."
log_saif [get_objects -r /tb_top_alu/UUT/*]
run 3300 ns
close_saif
```

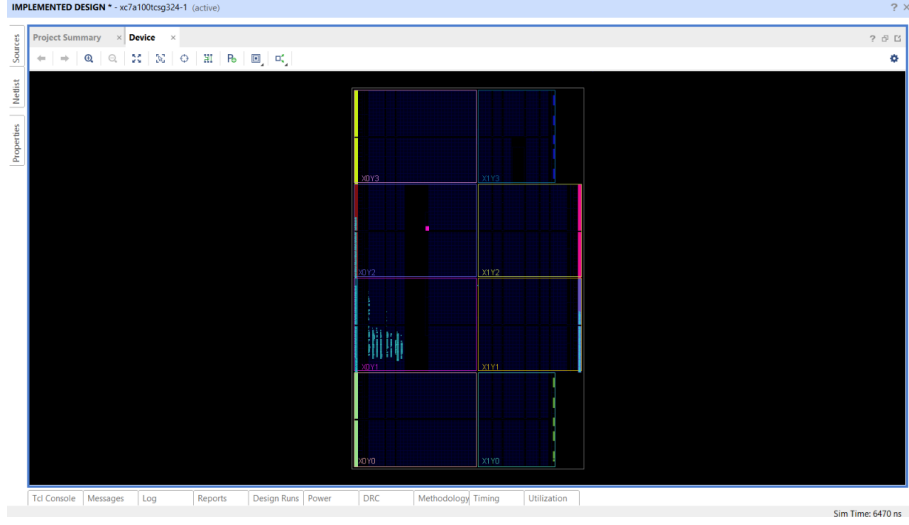
The recursive tracking flag (`-r`) commands the simulation engine to monitor not just the top-level ports, but all internal interconnections, flip-flops, and logic primitives within ‘U_ARITH’, ‘U_LOGIC’, and ‘U_SHIFT’. As the 3,300 ns stress test executes, every single logic transition is counted and compiled into the `sim_file.saif` matrix.

4 Physical Implementation & Floorplanning

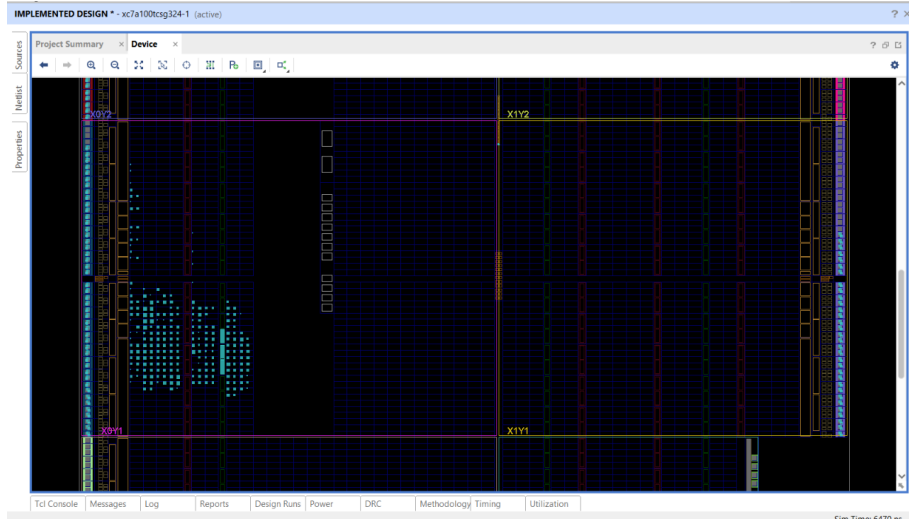
4.1 Silicon Floorplan and Physical Layout Analysis

To capture the macro-level spatial distribution of the hardware, the post-implementation physical layout was evaluated inside the Vivado Device Analysis engine. Floorplanning

analysis serves as a vital diagnostic step, proving how the logical hierarchies defined in VHDL are translated into localized physical layout coordinates on the silicon die.



(a) Full-chip macro floorplan layout.



(b) Magnified view of the active quadrant.

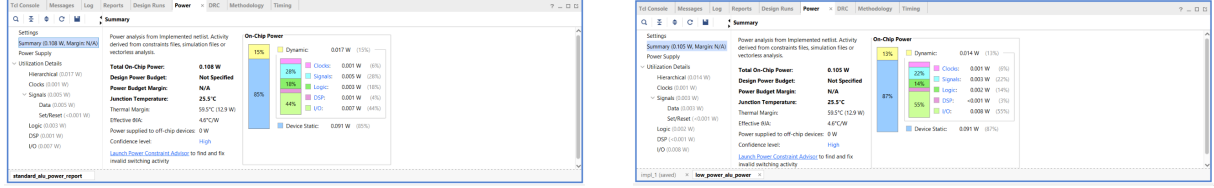
Figure 1: Post-Implementation device layout analysis.

As shown in Figure 1, the integrated design occupies a highly localized footprint, utilizing approximately 10% to 15% of the total available silicon real estate on the Artix-7 chip. The remainder of the fabric remains unallocated and dark. To perform a granular assessment of the routing density and gate distribution, the physical view was magnified directly into the active logic quadrant.

5 Experimental Results

5.1 Dynamic Consumption Analysis

Post-implementation power characterization was conducted by comparing the baseline standard ALU against the optimized operand-gated ALU. Both designs were analyzed under the identical 3,300 ns high stress test workload at an operating frequency of 50 MHz, utilizing 100% complete post-simulation SAIF back-annotation. Table 1 establishes the definitive data comparison matrix between the two design variants.



(a) Standard ALU Power Report.

(b) Low-Power ALU Power Report.

Figure 2: Post-Implementation Power Summaries showcasing the distribution of power dissipation inside the Vivado Design Suite.

Table 1: Post-Implementation Power Comparison Matrix at 50 MHz

Power Component	Standard ALU	Operand-Gated ALU	Delta
Total On-Chip Power	108 mW	105 mW	-2.78%
Device Static Power	91 mW	91 mW	0.00%
Dynamic Power	17 mW	14 mW	-17.65%
<i>Dynamic Breakdown:</i>			
Internal Signals	5 mW	3 mW	-40.00%
Internal Logic	3 mW	2 mW	-33.33%

Static power is fixed at 91 mW because it is strictly determined by the physical transistor leakage of the underlying 28 nm silicon substrate. RTL-level modifications cannot adjust static leakage; it remains active as long as the chip is powered. The true metric of design optimization must be tracked inside the **Total Dynamic Power** layer, which isolates the electrical energy consumed exclusively within the custom VHDL logic primitives and routing interconnect paths. Under the high stress workload, the baseline design dissipates 17 mW of dynamic power. Implementing the front-end operand-gating isolation multiplexers drives this metric down to 14 mW, achieving a noteworthy **17.65% reduction in dynamic power**.

This saving is explicitly validated by the internal breakdowns: *Internal Signals* power fell by 40.00% (from 5 mW to 3 mW) and *Internal Logic* dropped by 33.33% (from 3 mW to 2 mW). This proves that by successfully clamping unselected buses to zero, the switching activity factor (α) within the other sub-modules was completely eliminated, forcing their internal capacitances to maintain a zero-power static state.

5.2 Timing Characterization and the Power-Delay Trade-off

While operand gating provides a notable power reduction, it exposes a fundamental law of digital design: the **Power-Delay Trade-off**. To chart this performance penalty, the system was initially subjected to a 100 MHz clock tree constraint.

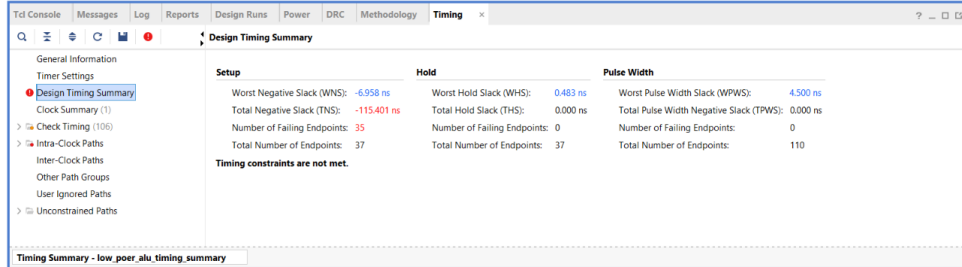


Figure 3: Timing Analysis Report at 100 MHz showing critical path setup time violations.

As captured in Figure 3, running at 100 MHz causes severe structural timing failures. The Gated Low-Power ALU yields a WNS of -6.958 ns. In a Gated ALU, the operands must first cross the internal transistors of the front-end gating multiplexers. This adds a localized layer of combinatorial gate propagation delay (t_{pd}) directly onto the entrance of an already deep critical path.

Furthermore, the control lines driving these gating cells must drive all 32 parallel bit lines simultaneously. This high fan-out architecture introduces localized parasitic capacitance, which slows down signal transition times and worsens setup time constraints.

The absolute delay path requires approximately 16.958 ns to complete its execution (10 ns (Target Period) + 6.958 ns (Worst Violation)). Therefore, scaling the system clock constraint to a standard **50 MHz operating frequency** provides an optimal timing margin.

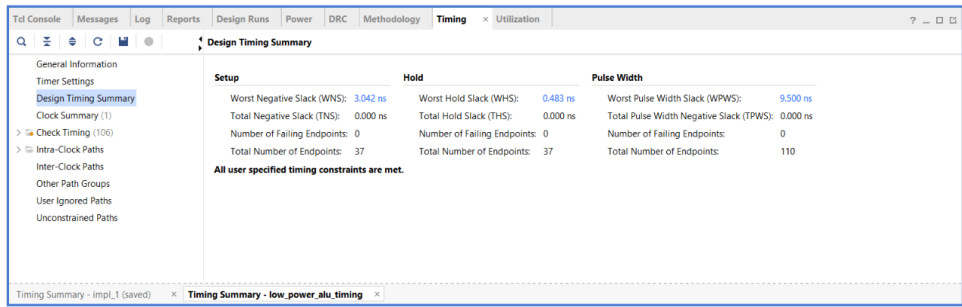


Figure 4: Final Post-Implementation Timing Summary showing successful Timing Closure at 50 MHz.

Figure 4 confirms successful **Timing Closure** of the optimized low-power design at 50 MHz. By widening the available calculation window to 20 ns, the signal fully stabilizes through the gating multiplexers and the hardware multiplier column with time to spare. The setup slack transitions into a completely stable, positive Worst Negative Slack of $+3.042$ ns. This positive slack mathematically guarantees absolute data integrity across all 16 operational sweeps.

6 Conclusion & Industrial Relevance

6.1 Summary of Achievements

This project successfully validates the application of microarchitectural operand gating as an effective, localized power-mitigation strategy for parallel computing structures. By embedding structural input-isolation multiplexers directly within the front-end routing pathways of a hierarchical 32-bit ALU, the design achieves a decisive framework that addresses power dissipation boundaries.

Under a high stress workload, the proposed architecture achieved a notable **17.65%** reduction in dynamic power compared to the standard configuration. Internal signal routing traces and combinational logic primitive power overhead dropped by 40.00% and 33.33%, respectively. This data confirms that dynamically clamping idle functional units to a static ground state effectively neutralizes redundant transistor switching activity ($\alpha \rightarrow 0$) without sacrificing operational throughput.

Furthermore, by comprehensively mapping the accompanying power-delay trade-off, full structural timing closure was safely established at an operating frequency of 50 MHz. The optimized netlist settled into a highly stable, positive Worst Negative Slack (WNS) of **+3.042 ns**, ensuring complete clock-tree synchronization and absolute data validation across all execution branches without requiring aggressive multi-stage register pipelining or incurring devastating area penalties.

6.2 Industrial Alignment

Beyond its immediate academic and laboratory results, the engineering methodology deployed throughout this project directly mirrors the rigorous verification and signoff protocols utilized by modern industrial VLSI design teams. In enterprise-grade hardware engineering, relying on vectorless tool estimations or arbitrary mathematical heuristics to predict power density profiles is entirely unacceptable. The implementation of an activity export pipeline, which includes running a post-implementation functional simulation to extract a 100% annotated Switching Activity Interchange Format (SAIF) matrix, represents the industry-standard paradigm for gate-level power signoff.

While this project specifically evaluated the microarchitecture on an FPGA fabric, the structural principles and verification models established here translate directly to Application-Specific Integrated Circuit (ASIC) development. In the production of modern mass-market System-on-Chip (SoC) microprocessors, mobile chipsets, and high-performance hardware accelerators, this exact front-end RTL gating logic is compiled and handed off to premium industrial signoff engines, such as **Synopsys PrimePower** or **Cadence Joules**.

These enterprise tools ingest the same SAIF signal-toggle profiles generated in this research to map localized thermal hot spots, analyze leakage-to-dynamic power ratios, and predict battery-drain metrics before committing a design to multi-million dollar silicon foundry tape-outs. Ultimately, this project demonstrates that low-power microarchitectural modifications, when paired with high-fidelity simulation-driven verification, provide hardware designers with a predictable, mathematically sound methodology to meet strict green-computing and thermal design budgets in deep-submicron silicon.